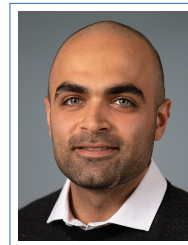


Hoomaan Hezave Hesar Maskan (1994-12-13)

Curriculum vitae

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📄 Hoomaan Hezaveh



[Click here for link to Google Scholar.](#)

Education and Employment History

- Sep 2013 – **B.Sc. in Electrical and Electronics Engineering**, *Iran University of Science and Technology*, Tehran, Iran, GPA: 15.99/20
- Sep 2017 – **M.Sc. in Telecommunication Engineering**, *Iran University of Science and Technology*, Tehran, Iran, GPA: 18.38/20
- Jan 2019 – Thesis: Sparse Signal Reconstruction using Blind Super-resolution with Arbitrary Sampling. I studied blind super-resolution of sparse signals using arbitrary sampling and the atomic lift, and proposed a semidefinite programming formulation using a matrix atomic norm. Supervisor: Prof. **M.H. Kahaei**.
- Aug 2020 – **WASP Doctoral Researcher and PhD in Mathematical Statistics**, *Umeå University*, Sweden
- Feb 2026 – Thesis: From Accelerated First-Order Methods to Structured Nonconvex Optimization: Analysis and Perspectives. I studied accelerated optimization methods through their continuous-time behaviour and structured nonconvex problems through difference-of-convex programming, with applications to machine learning. Supervisor: Assistant Prof. **Alp Yurtsever**. PhD completed in February 2026.
- Apr 2026 – **Postdoctoral Researcher in Optimization Theory**, *Uppsala University*, Sweden
- Present – Research on deterministic and stochastic optimization algorithms with applications to machine learning and operations research. Academic host: Associate Prof. **Prashant Singh**.

Internships

- Feb 2024 – **Research Intern**, *Pendulum Systems*, USA
- Nov 2024 – Prediction, resource allocation, end-to-end optimization and active learning. This internship took place during the doctoral period listed above.

Research Visits

- May 2023 – **Research Assistant**, *University of Edinburgh*, United Kingdom
- Jun 2023 – Host: Prof. Konstantinos C. Zygalakis. During this visit I finalized two research projects [10, 11] on accelerated first-order methods in optimization. I also worked on probabilistic circuits for fast Bayesian denoising with T. Klatzer and K. C. Zygalakis.
- Jun 2023 – **Research Visit**, *University of Oxford*, United Kingdom
- Jun 2023 – Host: Prof. Jared Tanner. During this visit I presented my research to Prof. Tanner.
- Mar 2025 – **Research Assistant**, *Technical University of Munich*, Germany
- May 2025 – Host: Prof. Suvrit Sra. During this visit I finalized a research project on blockwise difference-of-convex programming for structured nonconvex problems and initiated projects on portfolio optimization, bilevel programming and inverse optimization.

Computer skills

- Python Extensive modular based coding for training **neural networks** with newly developed algorithms and **probabilistic circuits**, **random forests** for data analysis, portfolio optimization, and Amazon's Last Mile Challenge, time series analysis, prediction, and optimization using **CVXpy**. I have course level experience with implementing Autoencoders, GANs, Language Models (LSTM), Bayesian Networks, and Graphical Models. Favorite Packages: *Pytorch*, *Pandas*, *NumPy*, *Scikit-learn*, *Matplotlib*, *CVXpy*
- MATLAB I have robotic implementation experience using MATLAB+Rduino, CVX optimization, Image Processing, Speech Processing, Fourier/Spectrum Analysis, Sparse Signal Recovery from noisy data

- Scala Distributed Training using Horovod-Spark Package. Course project experience (Scalable Data Science course)
- R Basic teaching experience using R
- Others Github, Microsoft Office, Latex, Atlassian, Jira, Bitbucket, AWS (self trained), Transformers (self trained), Docker (self trained)

Research Interests

Convex/Non-Convex Optimization, Bilevel Programming, Reinforcement Learning, Bandit, Difference of Convex Programming, Language Models, Training Algorithms, Accelerated algorithms, Stochastic methods in machine learning, Inverse problems, Sparse representation theory

Publications

- [1] Pouria Fatemi, Hoomaan Maskan, Suvrit Sra, and Peyman Mohajerin Esfahani. Tight generalization bounds for noiseless inverse optimization. In *The Fortieth Annual Conference on Neural Information Processing Systems*, 2026.
- [2] Pouria Fatemi*, Hoomaan Maskan*, Alp Yurtsever, and Suvrit Sra. The multi-block DC function class: Theory, algorithms, and applications. In *The Fortieth Annual Conference on Neural Information Processing Systems*, 2026 (*Main Authors).
- [3] Hoomaan Hezave, Milad Javadzadeh, and Mohammad Hossein Kahaei. Sparse signal reconstruction using blind super-resolution with arbitrary sampling. *IEEE Signal Processing Letters*, 27:615–619, 2020.
- [4] Hoomaan Hezaveh, Iman Valiulahi, and Mohammad Hossein Kahaei. Incorporation of prior knowledge into sparse time dispersive ofdm channel estimation via weighted atomic norm minimisation. *IET Communications*, 14(11):1704–1708, 2020.
- [5] Milad Javadzadeh Jirhandeh, Hoomaan Hezave, and Mohammad Hossein Kahaei. Super-resolution doa estimation for wideband signals using non-uniform linear arrays with no focusing matrix. *IEEE Wireless Communications Letters*, 11(3):641–644, 2021.
- [6] Aleksandr Karakulev*, Hoomaan Maskan*, Alp Yurtsever, and Prashant Singh. Non-euclidean coordinate dc programming. *Main Authors (Under Review)*, 2026.
- [7] Hoomaan Maskan, Sajad Daei, and Mohammad Hossein Kahaei. Demixing sines and spikes using multiple measurement vectors. *Signal Processing*, 203:108786, 2023.
- [8] Hoomaan Maskan, Paniz Halvachi, Suvrit Sra, and Alp Yurtsever. Randomized block coordinate dc algorithm. *EURO Journal on Computational Optimization*, page 100123, 2025.
- [9] Hoomaan Maskan, Yikun Hou, Suvrit Sra, and Alp Yurtsever. Revisiting frank-wolfe for structured nonconvex optimization. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [10] Hoomaan Maskan, Konstantinos C Zygalakis, Armin Eftekhari, and Alp Yurtsever. A unified model for high-resolution odes: New insights on accelerated methods. *arXiv preprint arXiv:2503.15136*, 2025 (Under Review).
- [11] Hoomaan Maskan, Konstantinos C Zygalakis, and Alp Yurtsever. A variational perspective on high-resolution odes. In *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
- [12] Xuhui Zhang, Hoomaan Maskan, Alp Yurtsever, and Suvrit Sra. Toward distributionally robust decision focused learning. *NeurIPS 2026 Second Workshop on MLxOR*, 2026 (Full version Under Review).

Talks

- June, 2026 **SIAM Conference on Optimization**, *Tight Generalization Bounds for Noiseless Inverse Optimization*, Edinburgh University, UK

- May, 2026 **ELLIIT Focus Period**, *The multi-block dc function class: Theory, algorithms, and applications*, Lund University, Sweden
- Feb, 2025 **Umeå University PhD Conference**, *Block Coordinate DC Programming*, Umeå University, Sweden
- Jun, 2024 **The 21st EUROpt Conference: continuous optimization**, *Revisiting Frank-Wolfe for Nonconvex Optimization*, Lund University, Sweden
- Dec, 2024 **New York University**, *A Variational Perspective on High-Resolution ODEs*
- Aug, 2023 **The 20th EUROpt Conference: continuous optimization**, *Revisiting High-Resolution ODEs for Faster Convergence Rates*, Corvinus University, Budapest
- Jun, 2023 **Imperial College London**, *A Variational Perspective on High-Resolution ODEs*
- Feb, 2023 **JSS Seminars-Umeå University**, *Revisiting High-Resolution ODEs for Faster Convergence Rates*

Teaching Experience

- 2026 **Teacher**, *Uppsala University*, Sweden, Computational Methods for Scientific Applications
- 2025 **Teacher**, *Umeå University*, Sweden, Basic Statistics for Engineers
- 2024 **Teacher**, *Umeå University*, Sweden, Digital Signal Processing (Transform Methods)
- 2023 **Teacher and Teacher Assistant**, *Umeå University*, Sweden, Digital Signal Processing (Transform Methods)
- 2021, 2022, 2023 **Teacher Assistant**, *Umeå University*, Sweden, Statistik för teknologer (Statistics for Engineering) 4 times
- 2019 **Teacher Assistant**, *Iran University of Science and Technology*, Tehran, T.A for Estimation Theory
- 2018 **Teacher Assistant**, *Iran University of Science and Technology*, Tehran, T.A for Discrete Signal Processing

Supervision Experience

- 2025 **Master Thesis**, *Technical University of Munich*, Germany, End-to-End portfolio allocation
- 2024 **Master Thesis**, *Scania*, Sweden, Weather impact on energy consumption of electric vehicles
- 2024 **Bachelor Student**, *Sharif University of Technology*, Iran, Block coordinate dc programming

Pedagogical Studies

- 2023 **Pedagogical Studies**, *Umeå University*, Sweden, New as University Teacher (Ny som lärare)

Achievements

- PostDoc **Funding**, *I was awarded WASP postdoctoral scholarship with a value of 1.1 M SEK from Wallenberg Foundation to conduct a 2 year PostDoctoral program at Inria Paris Center hosted by Directeur de Recherche Francis Bach.*
- PhD **Funding**, *I was awarded 20,000 SEK from Kempe Foundation to participate in the thirty-ninth Conference on Neural Information Processing Systems (NeurIPS 2025).*
- PhD **Funding**, *I was awarded 30,000 SEK from Wallenberg Foundation to participate in the thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023). Alongside this trip, I visited IBM and gave a talk on first-order accelerated methods in the New York University.*
- PhD **Funding**, *I was awarded 16,500 SEK from Kempe Foundation to participate in the thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023). Alongside this trip, I visited IBM and gave a talk on first-order accelerated methods in the New York University.*

Service to the community

- 2026 August **Reviewer**, *I served as a reviewer at TMLR*
- 2026 July **Organizer**, *I served as a session organizer at EUROPT 2026, Linz, Austria*
- 2026 July **Reviewer**, *I served as a reviewer at the journal of Advances in Computational Mathematics*
- 2026 April **Reviewer**, *I served as a reviewer at NeurIPS 2026*

2026 Feb **Reviewer**, *I served as a reviewer at ICML 2026*
2025 Nov **Reviewer**, *I served as a reviewer at ICLR 2026*
2025 Jul **Reviewer**, *I served as a reviewer in Journal of Optimization and Applications (JOTA)*
2025 Jun **Reviewer**, *I served as a reviewer at NeurIPS 2025*
2025 Jun **Reviewer**, *I served as a reviewer for Journal of Optimization and Applications*
2025 Feb **Reviewer**, *I served as a reviewer at ICML 2025*
2024 Nov **Reviewer**, *I served as a reviewer at ICLR 2025*
2024 Nov **Reviewer**, *I served as a reviewer at AISTATS 2025*
2024 Sep **Reviewer**, *I served as a reviewer at AAAI 2025*
2024 Aug **Reviewer**, *I served as a reviewer at NeurIPS 2024*
2024 Jan **Reviewer**, *I served as a reviewer at ICML 2024*
2023 Aug **Reviewer**, *I served as a reviewer at NeurIPS 2023*
2023 Nov **Reviewer**, *I served as a reviewer at ICLR 2024*
2022 Jan – **NTK**, *Member of PhD Student Council in Umeå University*
2025 Sep
2022 Sep – **WASP**, *Umeå University representative in WASP PhD Student Council*
2025 Sep

Languages

Persian Native
English Fluent
Swedish Fluent
German Basic

References

- Alp Yurtsever (PhD Advisor)
Assistant Professor at Umeå University , alp.yurtsever@umu.se
- Prashant Singh (PostDoc Mentor)
Associate Professor at Uppsala University, prashant.singh@it.uu.se
- Suvrit Sra (Collaborator)
Professor at Technical University of Munich (TUM), s.sra@tum.de, suvrit@mit.edu
- Konstantinos Zygalakis (Collaborator)
Associate Professor at Edinburgh University , kzygalak@exseed.ed.ac.uk
- Mohammad Hossein Kahaei (Master's Advisor)
Associate Professor at Iran university of Science and Technology, kahaei@iust.ac.ir